

LABORATORY REPORT

Course No: —

Course Title: —

Experiment No: 02

Name of the Experiment:

RC Circuit Test

Submitted by:

Submitted to:

—

Date of Performance: October 10, 2026

Date of Submission: October 17, 2026

Experiment No: 02

Name of the Experiment: RC Circuit Test

1 Introduction

2 Circuit Design

A variable DC voltage source is connected across the main terminals. A gate current is provided to trigger the device. The voltage is swept from negative to positive values.

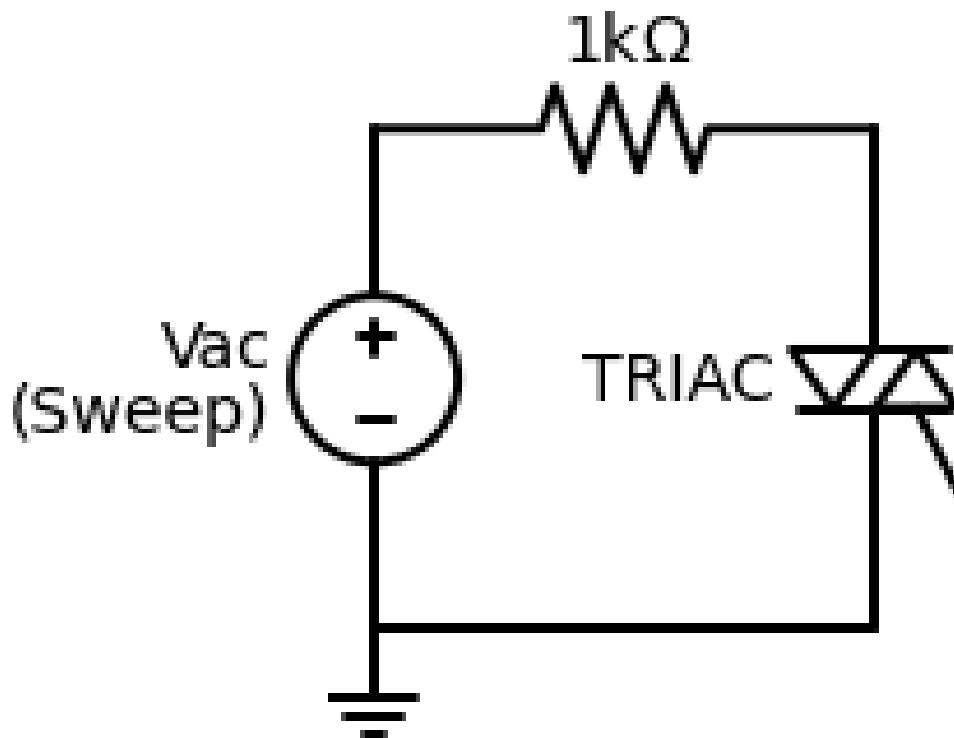


Figure 1: Experimental Circuit Diagram

3 Required Apparatus

- DC Power Supply (Variable) (Quantity: 1)
- Device Under Test (Quantity: 1)
- Resistor ($1k\Omega$) (Quantity: 1)
- Multimeter (Quantity: 2)

4 Procedure

1. Connect the circuit as per the experimental circuit diagram.
2. Apply a constant gate current I_G .

3. Vary the supply voltage from -15V to 15V.
4. Record the voltage and current.
5. Plot the characteristics.

5 Result

Voltage (V)	Current (mA)
-15.0	-15000.0
-11.7	-11700.0
-8.4	-0.0
-5.0	-5000.0
-1.7	-0.0
1.6	1600.0
5.0	-0.0
8.3	8300.0
11.7	-0.0
15.0	15000.0

Table 1: Simulated Data (Sample Points)

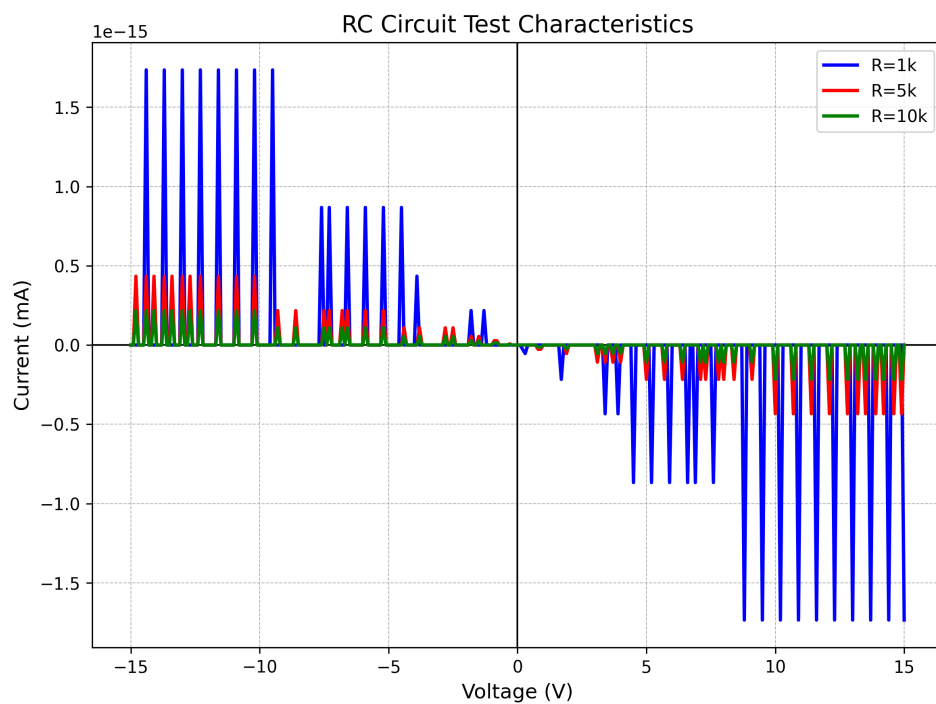


Figure 2: Simulated RC Circuit Test Curve

6 Discussion

7 Conclusion

8 References

1. Generated by LabGen Knowledge Hub API
2. Ngspice Simulation Data.